Journal of Mathematical Imaging and Vision
https://doi.org/10.1007/s10851-018-00870-z

Multiplicative Noise Removal Based on the Smooth Diffusion Equation

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Received: 8 October 2017 / Accepted: 24 December 2018
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Abstract
The multiplicative noise removal problem is of momentous significance in various image processing applications. In this paper, a nonlinear diffusion equation with smooth solution is proposed to remove multiplicative Gamma noise. The diffusion coefficient takes full advantage of two features of multiplicative noise image, namely, gradient information and gray level information, which makes the model has the ability to remove high level noise effectively and protect the edges. The existence of the solution has been analyzed by Schauder's fixed-point theorem. Some other theoretical properties such as the maximum principle are also presented in the paper. In the numerical aspect, the explicit finite difference method, fast explicit diffusion method, additive operator splitting method and Krylov subspace spectral method are employed to implement the proposed model. Experimental results show that the fast explicit diffusion method achieves a better trade-off between computational time and denoising performance, and the Krylov subspace spectral method gets better restored results in the visual aspect. In addition, the capability of the proposed model for denoising is illustrated by comparison with other denoising models.

Keywords Multiplicative noise removal · Diffusion equation · Smooth solution · Fast algorithm

1 Introduction

As a necessary preprocessing step, image denoising has received a lot of attention for many areas such as medical and aerospace. It is well known that noise usually contaminates image in additive, multiplicative and mixed forms. In this paper, we focus on removing multiplicative Gamma noise that appears often in the synthetic aperture radar (SAR) image [4,32] and we assume that the noise image satisfies the mathematical model as follows:

$$f = u\eta, \tag{1}$$

where f is the corrupted image, u the noise-free image and η the noise which follows the Gamma Law with mean equals to one. The probability distribution function of η reads:

$$g(\eta) = \frac{L^L}{\Gamma(L)}\eta^{L-1}\exp(-L\eta)\mathbf{1}_{\{\eta\geq 0\}},$$

where L is the number of looks and $\mathbf{1}_{\{\eta\geq 0\}}$ is the indicator function on the subset $\{\eta \geq 0\}$.

There are plenty of denoising methods, such as variational approaches [13,17,18,40,46], partial differential equations (PDEs) [21,22,25,48], filter approaches [6,23,28,30], wavelet approaches [2,7,12,22], among which the total variational (TV) methods obtain impressive results. The first TV-based approach to remove multiplicative Gaussian noise was introduced by Rudin et al. [35] in 2003. TV regularization was then used for multiplicative Poisson noise removal in [14]. Aubert and Aujol [5] proposed a variational model (AA) by the maximum a posteriori (MAP) estimation approach to remove multiplicative Gamma noise. The fidelity term of AA model was subsequently adopted in a convex model (OS) by Osher and Shi [37]. In Steidl and Teuber [39], a relation has been shown between the OS model and I-divergence model for multiplicative Poisson noise removal.

TV-based denoising favors solutions that are piecewise constant. This sometimes causes a staircasing effect in which noisy smooth regions are processed into piecewise constant regions, a phenomenon long observed in the literature [15]. Recently, high-order variational approaches [8,10,11], fractional-order variational approaches [47] and also nonlocal approaches [24] were proposed to avoid the staircasing

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Published online: 09 January 2019

effect and achieved some good results in additive noise removal.

PDEs have been known for a long time as an useful tool in image processing. One reason is that many variational problems or their regularized approximations can often be effectively computed from their Euler–Lagrange equations. In 1990, Perona and Malik [34] first developed a nonlinear filter to reduce additive Gaussian noise as follows:

$$\frac{\partial u}{\partial t} = \operatorname{div}(g(|\nabla u|)\nabla u),$$

where g is a smooth nonincreasing function with $g(0) = 1$, $g(x) \geq 0$ and $\lim_{x\to\infty} g(x) = 0$. The conduction coefficient in the diffusion process depends on the local characteristics $|\nabla u|$, which makes the model perform better than general linear filters. However, the denoising effect of PM model is unsatisfactory because it has a tendency to keep or even enhance the high level noise. In addition, it is difficult to study the model mathematically since the initial boundary problem is ill-posed.

Catté et al. [9] proposed the regularized PM (RPM) model to obtain a unique solution and better experimental performance by replacing $|\nabla u|$ by $|\nabla G_\sigma * u|$ in PM model. In 2002, Yu and Acton [45] introduced a nonlinear diffusion equation for speckle noise removal, which tends to produce dislocated and unsharp edges despite impressive noise reduction performance. Abd-Elmoniem et al. [1] presented a tensor-based anisotropic diffusion method for speckle reduction and coherence enhancement, but it has difficulties in retaining subtle features, such as small cysts and lesions in ultrasound images. Krissian et al. [27] extended the filter to a matrix anisotropic diffusion, allowing different levels of filtering across the image contours and in the principal curvature directions. In 2015, Zhou et al. [49] proposed a doubly degenerated (DD) diffusion model to remove multiplicative noise.

In this paper, we aim to propose a nonlinear diffusion model that can be used for multiplicative noise removal and also possesses better mathematical properties. In addition, the numerical methods for solving the model should be considered. Thus, in the following, a new nonlinear smooth diffusion model that inspired by the RPM model and DD model is proposed. The diffusion coefficient is defined in terms of the gray level and gradient of the image, so the diffusion in different regions is steerable. The Gaussian kernel function is also introduced in the diffusion coefficient, which leads to several desirable properties of our model. In numerical aspects, four different numerical methods are considered to the model implementation, including the explicit finite difference method, fast explicit diffusion method, additive operator splitting method and Krylov subspace spectral method. Explicit finite difference method is a traditional and easy approach to solve the diffusion equation but suffers from low computational speed. Others are three fast algorithms that have been used successfully for solving the diffusion problems in image processing.

This paper is organized as follows: In Sect. 2, we construct a new nonlinear smooth diffusion model based on the characteristics of multiplicative noise and two diffusion denoising models. The existence and uniqueness of the solution as well as other properties are presented in Sect. 3. In Sect. 4, four numerical methods are employed for the implementation of the proposed model. In Sect. 5, the comparisons of four numerical methods are concluded and the capability of our model is given by comparing with other models. Finally, the work is summarized in Sect. 6.

2 The Nonlinear Smooth Diffusion Model for Multiplicative Noise Removal

Among nonlinear diffusion models for image processing, the gradient of the image is often utilized for model construction. Using the ideas of the RPM model and DD model, we proposed a new nonlinear smooth diffusion model based on the gradient and gray value of the image for multiplicative Gamma noise removal. Before giving details about the proposed model, we first recall the RPM model and DD model.

2.1 Review of Two Models

In this paper, we assume Ω to be an open, connected, bounded subset of $\mathbb{R}^2$ with Lipschitz boundary $\partial\Omega$, and set $\Omega_T = \Omega \times (0, T)$ for some fixed time $T > 0$. We also denote by $\sup_\Omega u$ (resp., $\inf_\Omega u$) the $\operatorname{esssup}_\Omega u$ (resp., $\operatorname{essinf}_\Omega u$), if a function u belongs to $L^p(\Omega)$ for some $1 \leq p \leq \infty$.

The RPM model proposed by Catté et al. [9] is given as follows:

$$\begin{aligned}
&\frac{\partial u}{\partial t} = \operatorname{div}\left(\frac{1}{1+|\nabla u_\sigma|^2}\nabla u\right) \quad \text{in } \Omega_T,\\
&\frac{\partial u}{\partial \mathbf{n}} = 0 \quad \text{on } \partial\Omega \times (0, T),\\
&u = f \quad \text{on } \Omega \times \{t = 0\},
\end{aligned}$$

where $u_\sigma = G_\sigma * u$, $G_\sigma(x) = \frac{1}{2\pi\sigma^2}\mathrm{e}^{-\frac{\|x\|^2}{2\sigma^2}}$ and f is the noise image. The Gaussian convolution in the diffusion coefficient increases the robustness of the denoisng model by making the edge detector ignored of small details caused by additive noise. When $|\nabla u_\sigma|$ is large, the diffusion will be slow and edges will be kept. When $|\nabla u_\sigma|$ is small, the diffusion will be faster to effectively remove the noise in flat areas. Despite its advantages in removing additive Gaussian [42],

the RPM model cannot handle the multiplicative noise case, as illustrated in a numerical example in Sect. 2.3.

Generally, most mathematical methods for image denoising deal with the gray-scale image. Dong et al. [16] proposed a diffusion coefficient function $c(u)$, called gray level indicator, in a convex adaptive total variation model to remove multiplicative noise. It takes into account the gray level of noisy image and controls the speed of diffusion at different regions. As discussed in [16], the gray scale of image is useful information to steer the diffusion progress, especially for the multiplicative noise removal problem. Inspired by the gray level indicator, Zhou et al. [49] developed a doubly degenerated (DD) diffusion model to address the multiplicative noise removal problem

$$\begin{aligned}
&\frac{\partial u}{\partial t} = \operatorname{div}(g(u, |\nabla u|)\nabla u) \quad \text{in } \Omega_T,\\
&\frac{\partial u}{\partial \mathbf{n}} = 0 \quad \text{on } \partial\Omega \times (0, T),\\
&u = f \quad \text{on } \Omega \times \{t = 0\}.
\end{aligned}$$

The diffusion coefficient is given as follows:

$$g(u, |\nabla u|) = \frac{2|u|^\alpha}{M^\alpha + |u|^\alpha} \cdot \frac{1}{\left(1 + |\nabla u|^2\right)^{\frac{1-\beta}{2}}},$$

where $0 < \alpha,\ 0 < \beta < 1$ and $M = \sup_{x\in\Omega} u(x, t)$. $a(u) := \frac{2|u|^\alpha}{M^\alpha+|u|^\alpha}$ is a gray level indicator, which satisfies $0 \le a(u) \le 1$. $b(|\nabla u|) := \frac{1}{(1+|\nabla u|^2)^{(1-\beta)/2}}$ is devoted to an edge detection function. However, the authors did not perform a thorough theoretical analysis because of the degeneration in the edge detection function, i.e., $b(|\nabla u|) \to 0$ when $|\nabla u| \to \infty$, which brings difficulty in studying the theoretical properties of the denoising model.

2.2 The Proposed Model

In this paper, we propose a model that incorporates the strengths of the RPM and DD model for noise removal. The diffusion depends on the local information of the image, not only the gradient but also the gray value information. Moreover, the choice of the diffusion coefficient yields a model which we can show is theoretically sound.

To this end, the proposed model is given as follows:

$$\begin{aligned}
&\frac{\partial u}{\partial t} = \operatorname{div}\left(g(u_\sigma, |\nabla u_\sigma|)\nabla u\right) \quad \text{in } \Omega_T,\\
&\frac{\partial u}{\partial \mathbf{n}} = 0 \quad \text{on } \partial\Omega \times (0, T),\\
&u = f \quad \text{on } \Omega \times \{t = 0\},
\end{aligned} \tag{2}$$

where

$$g(u_\sigma, |\nabla u_\sigma|) = \left(\frac{u_\sigma}{M}\right)^\alpha \frac{1}{1 + |\nabla u_\sigma|^\beta},$$

where $M = \max_{x\in\Omega} |u_\sigma(x, t)|$ and f is the initial image, and α, β and σ are positive constants. Next, the motivation of our diffusion model is summarized.

As pointed in Zhou et al. [49], the mathematical model of multiplicative noise, $f = u\eta$, indicates that the noise level is related to the original image. The higher gray level of u is, the stronger noise will be. Thus, we should consider the information of gray level and introduce the $a(u_\sigma) := \left(\frac{u_\sigma}{M}\right)^\alpha$ as a gray level indicator to deal with noise in varying gray level regions. $a(u_\sigma)$ satisfies $0 \le a(u) \le 1$. On the other hand, the Gaussian convolution used in RPM model brings a lot of benefits, not only the robustness in denoising aspect, but also the well posedness in theoretical aspect. Therefore, we also introduce the Gaussian convolution in the following edge detector $b(|\nabla u_\sigma|) := \frac{1}{1+|\nabla u_\sigma|^\beta}$ with a varying parameter β. $b(|\nabla u_\sigma|)$ indicates the diffusion speed is related to the gradient of the image. The larger value is, the lower diffusion speed will be, resulting in better edge protection.

The introduced Gaussian kernel function in our diffusion coefficient brings a lot of advantages. It makes the diffusion model powerful in edges detection and avoids the degeneration arising in the DD model. In addition, it also leads to a smooth solution of our nonlinear diffusion model, which may not be expected in image denoising. We will show that our nonlinear smooth diffusion model is efficient in removing high level multiplicative Gamma noise and also protects the main edges of the image.

2.3 Comparing Our Model with RPM and DD Model

In this section, we compared our model with RPM and DD model, and our model should reap the benefits of both RPM and DD. Almost all the information of the image disappears under the influence of multiplicative noise, which makes the multiplicative noise removal task harder than additive Gaussian noise's. An experimental result (see Fig. 1) is given to illustrate that the RPM model cannot handle the multiplicative noise. As we can see, an unauthentic experimental result of RPM model is obtained near the high gray level regions. On the contrary, our model can protect edges and makes the restored image more clear for human eyes as shown in Fig. 1d. Our model employs the gray level indicator to denoise, which indicates introducing gray indicator in the diffusion coefficient is an optional way to remove the multiplicative noise.

Comparing our model with DD model, we give the following advantages of our model. The gray level indicator in our model is $a(u_\sigma) = \left(\frac{u_\sigma}{M}\right)^\alpha$, which acts as similar as $a(u) = \frac{2|u|^\alpha}{M^\alpha+|u|^\alpha}$ in the DD model. However, our gray indi-

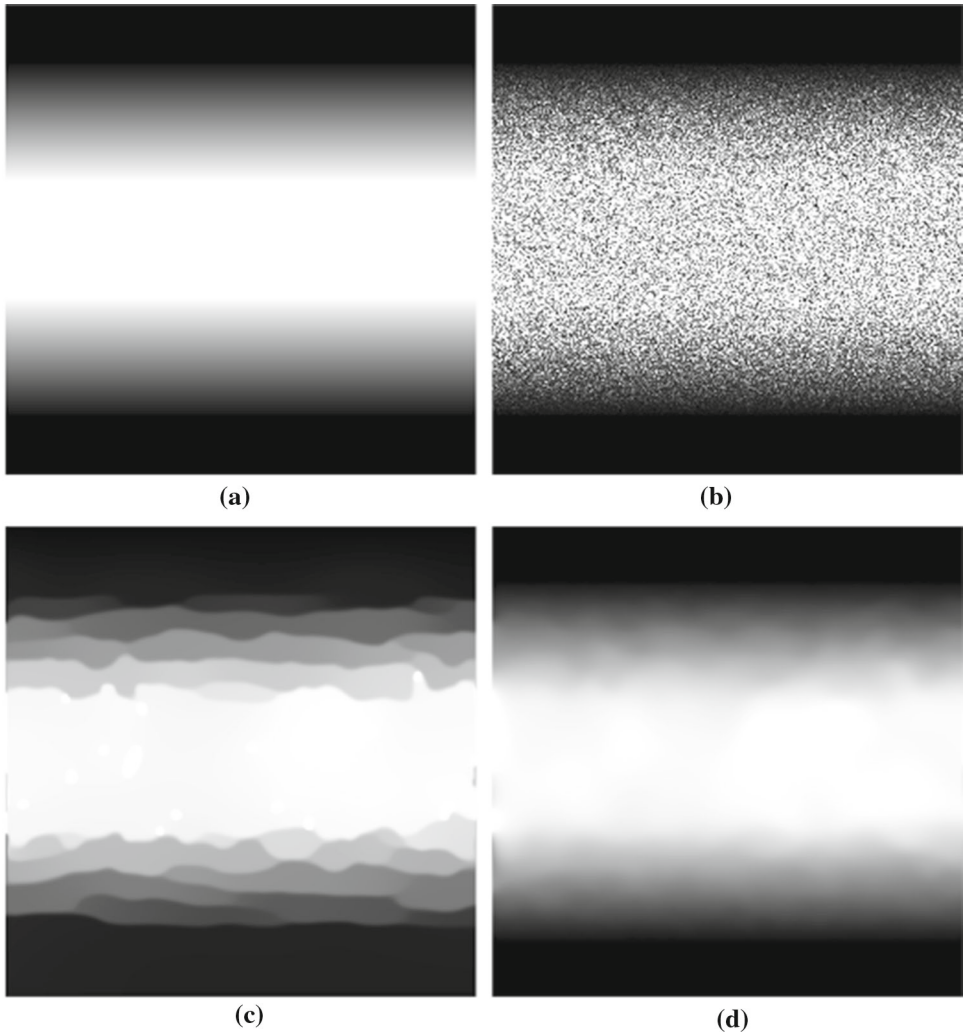

Fig. 1 Comparisons of RPM and our model in multiplicative noise removal. **a** Original image. **b** Multiplicative Gamma noise of mean 1 and standard deviation 0.5 ($L = 4$). **c** RPM with parameter $\sigma = 1$. **d** Our model with parameter $\sigma = 1,\ \alpha = 0.7$ and $\beta = 0.9$

cator is more concise and can be calculated more easily. DD model states that $a(u)$ makes the equation degenerated, when $u = 0$, while in our model we always assume $\inf_\Omega u > 0$ to avoid degeneration, which is an assumption also used in AA model and other denoising methods. The edge detection function used in DD model is $b(|\nabla u|) = \frac{1}{(1+|\nabla u|^2)^{(1-\beta)/2}}$, which also brings degeneration when $|\nabla u| \to \infty$. The edge detection function $b(|\nabla u_\sigma|)$ in our model avoids degeneration because of the Gaussian convolution. Furthermore, the parameter range of β in our model is $(0, +\infty)$, which is large than $\beta \in (0, 1)$ in DD model, making our model more flexible in parameter selection.

3 The Properties of Our Model

We study in this section the existence and uniqueness of the weak solution, its smoothness and other properties based on techniques introduced in Catté et al. [9].

3.1 Preliminaries

Let X denote a real Banach space with norm $\|\cdot\|_X$ and k denotes a positive integer. The definitions of spaces will be given first, which are essential in the following proofs.

The Sobolev space $W^{k,p}(\Omega)$ consisting of all locally summable function u such that for each multiindex α with $|\alpha| \le k$, $D^\alpha u$ exists in the weak sense and belongs to $L^p(\Omega)$. If $p = 2$, we usually write $H^k(\Omega) = W^{k,p}(\Omega)$ and define its norm to be

$$\|u\|_{H^k(\Omega)} := \left(\sum_{|\alpha| \le k} \int_\Omega |D^\alpha u|^2 \mathrm{d}x \right)^{1/2} < \infty.$$

$H^k(\Omega)$ is a Hilbert space. $H^k(\Omega)'$ denotes the dual space of $H^k(\Omega)$.

$L^p((0,T);X)$ denotes a space consisting of all strongly measurable functions $u:[0,T]\to X$ with

$$\|u\|_{L^p((0,T);X)}:=\left(\int_0^T \|u(t)\|_X^p \mathrm{d}t\right)^{1/p}<\infty$$

for $1<p<+\infty$, and

$$\|u\|_{L^\infty((0,T);X)}:=\sup_{0\le t\le T}\|u(t)\|_X<\infty.$$

The space $C([0,T];X)$ comprises all continuous functions $u:[0,T]\to X$ with

$$\|u\|_{C([0,T];X)}:=\max_{0\le t\le T}\|u(t)\|_X<\infty.$$

We then introduce $W(0,T)=\{u|u\in L^2((0,T);H^1(\Omega)),\ \frac{\partial u}{\partial t}\in L^2((0,T);H^1(\Omega)')\}$, and define its norm to be

$$\|u\|_{W(0,T)}:=\left(\int_0^T \|u(t)\|^2_{H^1(\Omega)}\mathrm{d}t+\int_0^T\left\|\frac{\partial u}{\partial t}\right\|^2_{H^1(\Omega)'}\mathrm{d}t\right)^{\frac{1}{2}}.$$

Then, $W(0,T)$ is a Hibert space [31].

For more details about the above spaces, we refer the reader to [3,20].

Definition A function $u\in W(0,T)\cap L^\infty((0,T);L^2(\Omega))$ is said to be a weak solution of problem (2) provided

$$\left\langle\frac{\partial u}{\partial t}(t),v\right\rangle_{H^1(\Omega)'\times H^1(\Omega)}+\int_\Omega g\big(u_\sigma,|\nabla u_\sigma|\big)\nabla u\nabla v\mathrm{d}x=0 \tag{3}$$

for each $v\in H^1(\Omega)$ and a.e. $t\in[0,T]$ and

$$u(0)=f. \tag{4}$$

Remark Referring to [20], Theorem 3 in Sect. 5.9.2, we have $u\in C([0,T];L^2(\Omega))$, and thus, the equality (4) makes sense.

3.2 Existence and Uniqueness of the Solution

In what next, we prove the existence and uniqueness of the weak solution by utilizing Schauder–Tikhonov fixed-point theorem [41] and weak convergence method [19]. Furthermore, we get a smooth solution based on the regularization method of the parabolic equation, Theorem 12.1 in [29].

Theorem 1 *Assume that $f\in L^2(\Omega)$ with $\inf_\Omega f>0$, then the problem (2) admits a weak solution u. Moreover, u belongs to $C^\infty(\bar\Omega\times(0,T])$.*

Proof 1. Let $w\in W(0,T)\cap L^\infty((0,T);L^2(\Omega))$ and set $w_\sigma=G_\sigma*w$. We denote the following problem by (Q_w)

$$\left\langle\frac{\partial u}{\partial t}(t),v\right\rangle_{H^1(\Omega)'\times H^1(\Omega)}+\int_\Omega g(w_\sigma,|\nabla w_\sigma|)\nabla u\nabla v\mathrm{d}x=0$$

for each $v\in H^1(\Omega)$ and a.e. $t\in[0,T]$ and

$$u(0)=f.$$

Since $w\in L^\infty((0,T);L^2(\Omega))$ and g, G_σ are C^∞, we can deduce that $g(w_\sigma,|\nabla w_\sigma|)\in L^\infty((0,T);C^\infty(\Omega))$ by $\nabla(w_\sigma)=\nabla G_\sigma*w$. Thus, there exists the constant $c_1\in\mathbb{R}^+$ depending only on g, f and G_σ, such that

$$0<c_1\le g(w_\sigma,|\nabla w_\sigma|)\le 1,\quad \text{a.e. in } \Omega_T \tag{5}$$

where $c_1>0$ follows from $\inf_\Omega f>0$.

2. According to (5) and the theory of classical parabolic equation in [31], there exists an unique solution u of (Q_w) in $W(0,T)$ such that

$$\|u\|_{L^2((0,T);H^1(\Omega))}\le C_1, \tag{6}$$

$$\|u\|_{L^\infty((0,T);L^2(\Omega))}\le\|f\|_{L^2(\Omega)}, \tag{7}$$

$$\left\|\frac{\partial u}{\partial t}\right\|_{L^2((0,T);H^1(\Omega)')}\le C_2, \tag{8}$$

for some constants C_1, C_2 depending on G_σ, g and f. These estimates lead us to introduce the subset W_0 of $W(0,T)$ defined by

$$\begin{aligned}W_0=\{&w\in W(0,T)|\\ &\|w\|_{L^2((0,T);H^1(\Omega))}\le C_1,\\ &\|w\|_{L^\infty((0,T);L^2(\Omega))}\le\|f\|_{L^2(\Omega)},\\ &\left\|\frac{\partial w}{\partial t}\right\|_{L^2((0,T);H^1(\Omega)')}\le C_2,\ w(0)=f\}.\end{aligned}$$

Then, from the definition of W_0, we know that W_0 is a nonempty, convex and weakly compact set of $W(0,T)$. Thus, there exists a sequence $\{w_j\}\in W_0$ and a function $w\in W_0$ so that

$$w_j\rightharpoonup w \text{ weakly in } L^2\Big((0,T);H^1(\Omega)\Big) \tag{9}$$

and

$$\frac{\partial w_j}{\partial t}\rightharpoonup\frac{\partial w}{\partial t} \text{ weakly in } L^2\Big((0,T);H^1(\Omega)'\Big). \tag{10}$$

We will hereafter abbreviate (9) and (10) by saying $w_j\rightharpoonup w$ weakly in W_0.

3. Based on the definition of (Q_w), we can define $U:W_0\to W_0$ is a mapping by setting $U(w)=u$. In order

to use Schauder–Tikhonov fixed-point theorem, we now need to prove that the mapping U is weakly continuous from W_0 to W_0, i.e., $\exists\ w_j \rightharpoonup w$ weakly in W_0 such that $U(w_j) \rightharpoonup U(w)$ weakly in W_0. Write $U(w_j) = u_j$. By using Rellich compactness theorem [19], (6) and (8), and passing to a subsequence if necessary, we have

$$\begin{aligned}
&u_j \rightharpoonup u \text{ weakly in } L^2\left((0,T); H^1(\Omega)\right),\\
&\frac{\partial u_j}{\partial t} \rightharpoonup \frac{\partial u}{\partial t} \text{ weakly in } L^2\left((0,T); H^1(\Omega)'\right),\\
&\frac{\partial u_j}{\partial x_i} \rightharpoonup \frac{\partial u}{\partial x_i} \text{ weakly in } L^2\left((0,T); L^2(\Omega)\right),\ i = 1,\ 2,\\
&u_j \to u \text{ in } L^2\left((0,T); L^2(\Omega)\right),\\
&u_j(0) \to u(0) \text{ in } L^2(\Omega).
\end{aligned}$$

Since $\{w_j\}$ is bounded in $L^\infty((0,T); L^2(\Omega))$, we deduce that $\{G_\sigma * w_j\}$ is bounded in $L^\infty((0,T); W^{1,\infty}(\Omega))$. Note that $\{\frac{\partial(G_\sigma * w_j)}{\partial t}\}$ is bounded in $L^2((0,T); H^1(\Omega)')$ from the definition of W_0. By utilizing Corollary 4 in [38] and the above conditions, we assert that

$$\begin{aligned}
&g(G_\sigma * w_j, |\nabla G_\sigma * w_j|) \to g(G_\sigma * w, |\nabla G_\sigma * w|)\\
&\text{uniformly in } C(\bar{\Omega} \times (0,T)).
\end{aligned}$$

Then, passing to the weak limit in (Q_{w_j}) yields $u = U(w)$. Moreover, by the uniqueness of the solution of (Q_w), the whole sequence $u_j = U(w_j)$ converges weakly to $u = U(w)$ in $W(0,T)$. Since $W(0,T)$ is contained in $L^2((0,T); L^2(\Omega))$, with compact inclusion, using Schauder–Tikhonov fixed-point theorem, we have $u \in W_0$ such that $u = U(u)$. Thus, the existence of the weak solution is proved.

4. Following Theorem 12.1 in [29] and the fact that u is a bounded weak solution of problem (2) that satisfies

$$0 < c_2 \le g(u_\sigma, |\nabla u_\sigma|) \le 1, \quad \text{a.e. in } \Omega_T, \tag{11}$$

for constant c_2 depending on f, g and G_σ, we can deduce that $u \in C^\infty(\bar{\Omega} \times (0,T])$. □

Theorem 2 *Assume that $f \in L^2(\Omega)$ with $\inf_\Omega f > 0$. Then, the solution u of (2) is unique.*

Proof Supposing that both u_1, u_2 are two solutions of (2) and setting $u_{\sigma 1} = G_\sigma * u_1$ and $u_{\sigma 2} = G_\sigma * u_2$, for a.e. $t \in [0,T]$, we have

$$\frac{\partial u_1}{\partial t} = \operatorname{div}\left(g\left(u_{\sigma 1}, |\nabla u_{\sigma 1}|\right) \nabla u_1\right), \tag{12}$$

$$\frac{\partial u_2}{\partial t} = \operatorname{div}\left(g\left(u_{\sigma 2}, |\nabla u_{\sigma 2}|\right) \nabla u_2\right), \tag{13}$$

$\frac{\partial u_1}{\partial \mathbf{n}} = 0$, $u_1(0) = f$, $\frac{\partial u_2}{\partial \mathbf{n}} = 0$ and $u_2(0) = f$. Then, combining (12) and (13), we obtain that

$$\begin{aligned}
&\frac{\partial(u_1 - u_2)}{\partial t} - \operatorname{div}\left(g\left(u_{\sigma 1}, |\nabla u_{\sigma 1}|\right)(\nabla u_1 - \nabla u_2)\right)\\
&\quad = \operatorname{div}\left(\left(g\left(u_{\sigma 1}, |\nabla u_{\sigma 1}|\right) - g\left(u_{\sigma 2}, |\nabla u_{\sigma 2}|\right)\right)(\nabla u_2)\right).
\end{aligned} \tag{14}$$

Multiplying Eq. (14) by $(u_1 - u_2)$, integrating over Ω, using the Neumann boundary condition and Hölder's inequality yields

$$\begin{aligned}
&\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega (u_1 - u_2)^2 \mathrm{d}x + c_2 \int_\Omega |\nabla u_1 - \nabla u_2|^2 \mathrm{d}x\\
&\quad \le \sup_{x\in\Omega}\left(g\left(u_{\sigma 1}, |\nabla u_{\sigma 1}|\right) - g\left(u_{\sigma 2}, |\nabla u_{\sigma 2}|\right)\right) \cdot\\
&\qquad \left(\int_\Omega |\nabla u_1 - \nabla u_2|^2 \mathrm{d}x\right)^{\frac{1}{2}} \left(\int_\Omega |\nabla u_2|^2 \mathrm{d}x\right)^{\frac{1}{2}},
\end{aligned} \tag{15}$$

where c_2 is the constant in (11). Since G_σ and g are C^∞, there exists a constant C depending only on G_σ, g and f such that

$$\begin{aligned}
&\sup_{x\in\Omega}\left(g\left(u_{\sigma 1}, |\nabla u_{\sigma 1}|\right) - g\left(u_{\sigma 2}, |\nabla u_{\sigma 2}|\right)\right)\\
&\quad \le C\|u_1 - u_2\|_{L^2(\Omega)},
\end{aligned} \tag{16}$$

Then, employing Young's inequality and (16) into (15), we have

$$\begin{aligned}
&\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega (u_1 - u_2)^2 \mathrm{d}x + c_2 \int_\Omega |\nabla u_1 - \nabla u_2|^2 \mathrm{d}x\\
&\quad \le \frac{C\varepsilon}{2}\int_\Omega |\nabla u_1 - \nabla u_2|^2 \mathrm{d}x\\
&\qquad + \frac{C}{2\varepsilon}\int_\Omega |u_1 - u_2|^2 \mathrm{d}x \int_\Omega |\nabla u_2|^2 \mathrm{d}x.
\end{aligned}$$

By setting $\varepsilon = 2c_2/C$, we deduce that

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega (u_1 - u_2)^2 \mathrm{d}x \le \frac{C^2}{2c_2}\int_\Omega (u_1 - u_2)^2 \mathrm{d}x \int_\Omega |\nabla u_2|^2 \mathrm{d}x. \tag{17}$$

Thus, the uniqueness of the solution follows from (17), Gronwall's inequality and $u_1(0) = u_2(0)$. □

3.3 Maximum Principle

This section develops the maximum principle for the proposed nonlinear diffusion model. The result based upon the weak solution is consequently almost everywhere in character, and it can be further extended into point-wise sense by the smoothness of the solution.

Theorem 3 *Assume that $f \in L^\infty(\Omega)$ with $\inf_\Omega f > 0$ and u is the solution of (2), then*

$$0 < \inf_\Omega f \le u \le \sup_\Omega f \quad \text{in } \bar{\Omega} \times (0, T].$$

Proof Define

$$(u-N)_+ = \begin{cases} u - N & \text{if } u - N \ge 0 \\ 0 & \text{otherwise,} \end{cases}$$

where $N = \sup_{x\in\Omega} f$. Replacing v by $(u-N)_+$ in (3) to get

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\int_{\{\Omega;u\ge N\}}(u-N)^2\mathrm{d}x + \int_{\{\Omega;u\ge N\}}\left(\frac{u_\sigma}{M}\right)^\alpha \frac{|\nabla u|^2}{1+|\nabla u_\sigma|^\beta}\mathrm{d}x = 0. \tag{18}$$

The second term of the left-hand side in (18) satisfies

$$0 \le \int_{\{\Omega;u\ge N\}}\left(\frac{u_\sigma}{M}\right)^\alpha \frac{|\nabla u|^2}{1+|\nabla u_\sigma|^\beta}\mathrm{d}x,$$

which yields that

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\{\Omega;u\ge N\}}(u-N)^2\mathrm{d}x \le 0. \tag{19}$$

Therefore, $\int_{\{\Omega;u\ge N\}}(u-N)^2\mathrm{d}x$ is decreasing for $t \in [0, T]$, since $u \in C([0, T]; L^2(\Omega))$ and (19). Then, from (19) and

$$\left(\int_{\{\Omega;u\ge N\}}(u-N)^2\mathrm{d}x\right)\Big|_{t=0} = 0$$

and

$$\int_{\{\Omega;u\ge N\}}(u-N)^2\mathrm{d}x \ge 0,$$

we have

$$\int_{\{\Omega;u\ge N\}}(u-N)^2\mathrm{d}x = 0, \quad \text{for } t \in [0, T].$$

Hence, we obtain that $u \le N = \sup_{x\in\Omega} f$ for $t \in [0, T]$ and a.e. $x \in \Omega$. Due to the smoothness of u for $t > 0$, we prove the assertion

$$u \le \sup_\Omega f \quad \text{in } \bar{\Omega} \times (0, T].$$

Replacing v by $(u+N)_+$ in (3), a similar argument yields that $\inf_\Omega f \le u$ in $\bar{\Omega} \times (0, T]$. □

Remark The maximum principle guarantees that overshoots and undershoots cannot appear in the resorted image and provides an important stability criterion in the numerical aspect [44].

3.4 Other Properties of the Solution

Other properties such as long-time behavior of the solution will be presented in this section.

Theorem 4 *Assume that $f \in L^\infty(\Omega)$ with $\inf_\Omega f > 0$. Then, the solution u of (2) provides*

$$\lim_{t\to\infty}\|u(t) - \bar{f}\|_{L^2(\Omega)} = 0,$$

where $\bar{f} = \frac{1}{|\Omega|}\int_\Omega f\mathrm{d}x$ and $|\Omega|$ denotes the Lebesgue measure of Ω.

Proof Multiplying $(u-\bar{f})$ with the first equation in (2), integrating over Ω and using the Neumann boundary condition, we deduce that

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega(u-\bar{f})^2\mathrm{d}x + \int_\Omega|\nabla u|^2 g(u_\sigma, |\nabla u_\sigma|)\mathrm{d}x = 0. \tag{20}$$

Employing (11) in Theorem 1 and Poincaré's inequality into (20) implies

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega(u-\bar{f})^2\mathrm{d}x + c_3\|u-\bar{f}\|^2_{L^2(\Omega)} \le 0, \tag{21}$$

where c_3 depends on G_σ, g and f and Ω. Set $y(t) = \int_\Omega(u-\bar{f})^2\mathrm{d}x$. Equation (21) can be rewritten as $y'(t) + c_3 y(t) \le 0$. Thus, there exists a constant $t_0 > 0$ such that $y(t) = 0$ for $t \ge t_0$. Therefore, we have

$$\lim_{t\to\infty}\left\|u(t) - \bar{f}\right\|_{L^2(\Omega)} = 0.$$

□

Next, we show that the restored image u has the same mean as the initial image f, which satisfies the characteristics of multiplicative noise η.

Theorem 5 *Assume that $f \in L^\infty(\Omega)$ with $\inf_\Omega f > 0$ and u be the solution of (2). Then*

$$\int_\Omega f\mathrm{d}x = \int_\Omega u\mathrm{d}x \quad \text{for } t \in [0, T].$$

Proof Integrating the first equation in (2) over Ω leads to

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega u\mathrm{d}x = \int_\Omega \mathrm{div}\left(\left(\frac{u_\sigma}{M}\right)^\alpha \frac{\nabla u}{1+|\nabla u_\sigma|^\beta}\right)\mathrm{d}x. \tag{22}$$

Then, using Green's formulas and the Neumann boundary condition, we know that

$$\begin{aligned}&\int_\Omega \operatorname{div}\left(\left(\frac{u_\sigma}{M}\right)^\alpha \frac{\nabla u}{1+|\nabla u_\sigma|^\beta}\right)\mathrm{d}x\\ &=\int_{\partial\Omega}\left(\frac{u_\sigma}{M}\right)^\alpha \frac{\nabla u}{1+|\nabla u_\sigma|^\beta}\cdot \mathbf{n}\mathrm{d}s=0.\end{aligned} \tag{23}$$

From (22) and (23), we have $\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega u\mathrm{d}x=0$ and the assertion $\int_\Omega f\mathrm{d}x=\int_\Omega u\mathrm{d}x$ for $t\in[0,T]$ follows. □

4 Numerical Algorithms

In this section, we presented four numerical methods including the explicit finite difference method (EFDM), fast explicit diffusion (FED) method, additive operator splitting (AOS) method and Krylov subspace spectral (KSS) method for the implementation of our model.

4.1 Explicit Finite Difference Method

Some notations are given for the following numerical scheme.
$x_i=i,\ y_j=j,\ i=0,1\ldots I,\ j=0,1\ldots J,$
where $I,\ J$ are the grid numbers of the image.
$t_n=n\Delta t,\ n=0,1\ldots,$
$u^n_{i,j}=u(x_i,y_j,t_n),$
$u^0_{i,j}=f_{i,j}=f(x_i,y_j),$
$u^n_{i,0}=u^n_{i,1},\ u^n_{0,j}=u^n_{1,j},$
$u^n_{i,J}=u^n_{i,J-1},\ u^n_{I,j}=u^n_{I-1,j},$
$\Delta^x_\pm u_{i,j}=\pm(u_{i\pm1,j}-u_{i,j}),$
$\Delta^y_\pm u_{i,j}=\pm(u_{i,j\pm1}-u_{i,j}),$
$m(a,b)=\frac{\mathrm{sgn}(a)+\mathrm{sgn}(b)}{2}\min(|a|,|b|).$

By following a standard explicit finite difference scheme proposed in [36], the EFDM algorithm for the proposed model is given as follows:

Algorithm 1 EFDM for the proposed model.
Set $u^0_{i,j}=f$.
For $n=0,1,\cdots$
$\quad Gu^n=G_\sigma * u^n_{i,j}.$
$\quad a^n_{i,j}=\left(\frac{Gu^n}{\max(Gu^n)}\right)^\alpha.$
$\quad b^{x,n}_{i,j}=1+((\Delta^x_+(Gu^n))^2+(m(\Delta^y_+(Gu^n)),\Delta^y_-(Gu^n)^2))^{\frac{\beta}{2}}.$
$\quad b^{y,n}_{i,j}=1+((\Delta^y_+(Gu^n))^2+(m(\Delta^x_+(Gu^n)),\Delta^x_-(Gu^n)^2))^{\frac{\beta}{2}}.$

$$\quad u^{n+1}_{i,j}=u^n_{i,j}+\Delta t\cdot\left(\Delta^x_-\left(a^n_{i,j}\cdot\frac{\Delta^x_+u^n_{i,j}}{b^{x,n}_{i,j}}\right)+\Delta^y_-\left(a^n_{i,j}\cdot\frac{\Delta^y_+u^n_{i,j}}{b^{y,n}_{i,j}}\right)\right).$$

End

4.2 Fast Explicit Diffusion Method

FED has been introduced by Grewenig and Weikert [43] for the implementation of anisotropic diffusion filter. Based on the explicit finite difference scheme, the main idea of this method is to replace originally fixed time steps by cycles of varying time steps. Most of the variant time steps do not follow traditional stability requirement in the explicit finite difference scheme. By establishing the connection between linear symmetric filters and explicit homogeneous diffusion schemes in one dimension, the insight of FED approach for parabolic problems was given in [43].

The explicit scheme for the proposed diffusion equation can be rewritten as

$$U^{n+1}=\left(E+\Delta t A(U^n)\right)U^n,$$

where E denotes a unit matrix and U^n is a vector containing the values at each pixel at n^{th} time level. Supposing that the nth step iterative coefficient matrix is P, then the cyclic acceleration strategy of FED is concluded as follows:

$$\begin{aligned}&U^{n+1,0}=U^n,\\ &U^{n+1,i+1}=\left(E+\Delta t_i P\right)U^{n+1,i},\\ &\Delta t_i=\frac{2}{\mu_{\max}(P)}\cdot\frac{1}{2\cos^2(\pi\cdot\frac{2i+1}{4k+2})},\quad i=0,\ldots,k-1.\end{aligned}$$

Δt_i is the time step of the cyclic iteration, and $\mu_{\max}(P)$ denotes the spectral radius of matrix P. The total time of a complete cycle to get $U^{n+1}=U^{n+1,k}$ satisfies

$$\theta_k=\frac{2}{\mu_{\max}(P)}\cdot\frac{k^2+k}{3}. \tag{24}$$

Then, the number of cyclic iterations k can be determined by total diffusion time T and the number of outer-iterations M.

Algorithm 2 FED for the proposed model.
Define $U^{1,0}=f$ *and set* $T,\ M$.
Initialization:
Find the cycle length k according to (24).
Compute $\Delta t_1\cdots\Delta t_{k-1}$ *according to*
$\Delta t_i=\frac{2}{\mu_{\max}(P)}\cdot\frac{1}{2\cos^2(\pi\cdot\frac{2i+1}{4k+2})}.$
Rearrange Δt_i *to* $\hat{\Delta t}_i$ *according to Leja ordering for numerical stability.*
Filtering Loop:
For $n=0,\ \cdots,\ M-1$
$\quad P=A(U^n)$
$\quad$ *For* $i=1,\ \cdots,\ k$
$\qquad U^{n,i+1}=(E+\Delta\hat{t}_i P)U^{n,i}.$
$\quad$ *End*
End

4.3 Additive Operator Splitting Method

The AOS method proposed by Weikert et al. [44] can also be utilized for the implementation of the proposed model. Let u^n denote the numerical solution at time t_n. $t_n = n\Delta t$ for the time step Δt. For two-dimensional data u^n, we do one-dimensional semi-implicit discretization for rows and columns of u^n separately as follows:

$$\left(E - 2\Delta t A_x^n\right)u_1^{n+1} = u^n, \tag{25}$$

$$\left(E - 2\Delta t A_y^n\right)u_2^{n+1} = u^n. \tag{26}$$

Then, taking the average of the two results as a full iteration result, i.e.,

$$u^{n+1} = \frac{1}{2}\left(u_1^{n+1} + u_2^{n+1}\right).$$

As for (25), the scheme for one-dimensional diffusion of the ith row can be written as

$$\begin{aligned} u_{i,j}^{n+1} = u_{i,j}^n + \Delta t \Bigg(& \frac{g_{i,j}^n + g_{i,j+1}^n}{2} u_{i,j+1}^{n+1} \\ & + \frac{g_{i,j}^n + g_{i,j-1}^n}{2} u_{i,j-1}^{n+1} \\ & - \frac{g_{i,j+1}^n + g_{i,j-1}^n + 2g_{i,j}^n}{2} u_{i,j}^{n+1} \Bigg), \\ j = 1, \ldots, J. & \end{aligned} \tag{27}$$

Writing (27) in matrix form, we have

$$\left(E - 2\Delta t A_{x,i}^n\right) u_{1,i}^{n+1} = u_i^n, \tag{28}$$

where $A_{x,i}^n$ is a $J \times J$ dimensional tridiagonal matrix. Then, write (28) for $i = 1, 2, \ldots, I$ as

$$\left(E - 2\Delta t A_x^n\right) u_1^{n+1} = u^n. \tag{29}$$

Finally, we get u_1^{n+1} by employing Thomas algorithm in (29).

Algorithm 3 AOS for the proposed model.
Define $u^0 = f$.
For $n = 0, 1, \ldots$
  For $i = 1, \ldots, I$
    Calculate $E - 2\Delta t A_{x,i}^n$.
    Using Thomas algorithm solves
    $(E - 2\Delta t A_x^n)u_1^{n+1} = u^n$.
  End
  For $j = 1, \ldots, J$
    Calculate $E - 2\Delta t A_{y,j}^n$.
    Using Thomas algorithm solves
    $(E - 2\Delta t A_y^n)u_2^{n+1} = u^n$.
  End
  $u^{n+1} = \frac{1}{2}(u_1^{n+1} + u_2^{n+1})$.
End

In general, time step Δt should be taken less than 5 to ensure the numerical accuracy of the AOS scheme.

4.4 Krylov Subspace Spectral Method

1-Node block KSS method [26] is applied here to implement our model. We now consider the proposed model of the form

$$\frac{\partial u}{\partial t} = \operatorname{div}\left(g\left(u_\sigma, |\nabla u_\sigma|\right)\nabla u\right) \tag{30}$$

on $[0, 1]^2$, with initial data f and periodic boundary conditions. For convenience, we consider the square image in KSS method and N denotes the number of grid points. From the discrete Fourier transform (DFT) $\mathcal{F}_{N,N}$, the discretization of two-dimensional differentiation operator $D_{z;N,N}$ can be obtained, where $z = x, y$. Then, Eq. (30) can be rewritten as an ODE form

$$\frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} = L_{N,N}\mathbf{u},$$

$$L_{N,N} = \sum_{z=x,y} D_{z;N,N} G_{N,N} D_{z;N,N},$$

where $G_{N,N}$ is a diagonal matrix consisting of the values at the grid points of $g(u_\sigma, |\nabla u_\sigma|)$.

The average of a function $f(x)$ on $[0, 1]^2$ is denoted by $\bar{f}$. $\hat{e}_\omega$ is a discretization of $\exp[2\pi i\omega x]$. Operator L is a linearization of the operator on the right-hand side of (30), defined by $Lu = \operatorname{div}\left(g\left(u_\sigma^n, |\nabla u_\sigma^n|\right)\nabla u\right)$. The algorithm of 1-node block KSS method for the proposed model is given as follows:

Algorithm 4 1-node block KSS method for the model.
Define $\hat{u}^n = \mathbf{fft}(u^n)$, $v = Lu^n$, $\hat{v} = \mathbf{fft}(v)$ and $u_\omega^n = u^n - \hat{e}_\omega \hat{e}_\omega^H u^n$.
  For each ω *do*
    $a_{11} = -(2\pi\omega)^2\bar{g}$
    $a_{12} = (\hat{v}(\omega) - a_{11}\hat{u}^n(\omega))/\|u_\omega^n\|_2$
    $a_{22} = (< u^n, v > + a_{11}\hat{u}^n(\omega)|^2$
      $- 2Re(\hat{u}^n(\omega)\hat{v}(\omega)))/\|u_\omega^n\|_2^2$
    $r = a_{22} - a_{11},\ b_{12} = |a_{12}|^2$
    $d = \sqrt{r^2 + 4b_{12}}, s = \operatorname{sgn}(r)$,
    $t = -2s/(sa_{21} + d)$
    $\omega_1 = 1/\sqrt{1 + b_{12}t^2},\ \omega_2 = \overline{a_{12}}t\omega_1$
    $\lambda_1 = a_{11} + b_{12}t,\ \lambda_2 = a_{22} - b_{12}t$
    $e_1 = \exp(\lambda_1\Delta t),\ e_2 = \exp(\lambda_2\Delta t)$
    $\hat{u}^{n+1}(\omega) = (|\omega_1|^2 e_1 + |\omega_2|^2 e_2)\hat{u}^n(\omega) + (e_1 - e_2)\omega_1\overline{\omega_2}\|u_\omega^n\|_2$
  End
$u^{n+1} = \mathbf{ifft}(\hat{u}^{n+1})$

In the 1-node block KSS method, discrete Fourier transform is used for the spectral discretization of the differential operator and Gauss quadrature rule is employed to approximate the bilinear type for the calculation of the Fourier coefficients. Theoretical analysis of this numerical method has been done by Palchak et al. [33] in recent years.

4.5 Analysis of Four Algorithms

The algorithms of four numerical methods have been summarized in the last section. Now, we provide the analysis of four numerical methods as follows:

Explicit finite difference method is a common way used to discretize nonlinear diffusion equations, which possesses a higher accuracy when compared with several methods, such as semi-implicit [44]. However, the numerical stability of EFDM requires very small time steps, for example, $\Delta t \leq \frac{1}{4}$ for two-dimensional linear problems. Considering the poor experimental efficiency of this method, we use other three fast algorithms for the model implementation.

By changing the fixed small time steps into varying time steps, FED is, in fact, a variant of explicit finite difference scheme, which is applicable in both diffusion and elliptic PDEs in image processing. As an unconditionally stable semi-implicit method, AOS method can speed up iterations by choosing large time steps. However, this method is only available to solve diffusion equations. Compared with AOS scheme, FED scheme has less computational error when both methods are discretized with first order in time, resulting in less computing time to reach the same error. In Weickert et al. [43], the authors provided that FED and AOS methods are two fast algorithms that are broadly applicable, easy to implement and well suited for parallel computing.

Besides the above two fast algorithms [43], KSS method is an efficient numerical method that is applicable for nonlinear diffusion models in image denoising and deblurring. Despite complicated theoretical analysis and numerical implementation, KSS method provides, as concluded by the authors in [22], a "best-of-both worlds" compromise between the efficiency of explicit time-stepping methods and the stability of implicit methods, which allows high-order accuracy in time and favorable stability properties.

In image processing, good experimental results are preferred than approximating diffusion equations as precisely as possible. Thus, we are more interested in the numerical methods that have better experimental results and efficiency. Four numerical methods are used to test our model, and their advantages and disadvantages will be presented quantitatively and qualitatively in Sect. 5.

5 Experimental Results

In this section, we first compare the four numerical algorithms, and give the selections of parameters α and β. Then, we compare quantitatively and qualitatively our model with some popular methods such as AA, OS and DD. All the tests are performed under Windows 10 and MATLAB R2014a, running on a laptop with an Intel(R) i3TM CPU (2.53GHz) and 2GB memory. The Gamma noise is generated by the MATLAB built-in function "gamrnd." In all tests, the initial value is set to be the observed image f. Different quantitative parameters are used to measure the quality of experimental results, including the peak signal-to-noise ratio (PSNR) and the mean absolute-deviation error (MAE) values

$$\mathrm{PSNR} = 10\log_{10}\frac{IJ|\max u_0 - \min u_0|^2}{\|u-u_0\|_{L^2}^2},$$

$$\mathrm{MAE} = \frac{\|u-u_0\|_{L^1}}{IJ},$$

where $|\max u_0 - \min u_0|$ gives the gray-scale range of original image and I, J denote the length and width of the image.

5.1 Comparisons of Algorithms 1–4

We tested four images, Lena (256×256), Cameraman (256×256) and two aerial images Nimes (400×400) and Hdr (512×512) for four algorithms, which are distorted by multiplicative Gamma noise with mean 1 and $L \in \{1, 4, 10\}$. The two aerial images are selected in numerical experiments because of their similarities with real SAR images.

The PSNR, MAE and CPU (in seconds) of the restored images for four algorithms are concluded in Table 1. The corresponding parameters are chosen such that the PSNR/MAE are optimal for all numerical algorithms. As we can see, each algorithm shows the numerical stability as noise level increases. Overall speaking, for various images including Lena, Cameraman and aerial images, each algorithm has the ability to get good experimental results. Furthermore, it shows that the FED method is the most competitive in terms of both PSNR/MAE and CPU time (s) from Table 1. In the visual aspect, KSS provides better resorted results in the details of the image, which can be seen from Fig. 2.

5.2 Parameters Selection

The selections of parameters α, β are given based on the above experiments. As different numerical algorithms lead to different calculation errors for the same equation, we vary the parameters to achieve the best experimental results for each algorithm. Table 2 summarizes the optimal values of α and β for all the algorithms for our model when the maximum PSNR and minimum MAE are obtained. As we can see from

Table 1 Comparisons of algorithms 1–4 for our model, where $\sigma = 1$

		EFDM			FED			AOS			KSS		
		PSNR	MAE	CPU(s)	PSNR	MAE	CPU(s)	PSNR	MAE	CPU(s)	PSNR	MAE	CPU(s)
$L = 1$	Lena	21.64	12.91	2223.58	**21.67**	**12.90**	**55.61**	21.59	12.92	171.07	21.62	13.27	257.22
	Cameraman	21.86	10.75	2848.58	**21.86**	**10.74**	**78.11**	21.66	10.86	359.05	21.81	11.47	288.49
	Nimes	23.84	11.10	51.93	23.82	11.09	10.97	**23.86**	**11.06**	**9.59**	23.80	11.29	11.57
	Hdr	**20.58**	**16.89**	191.67	20.58	**16.90**	17.55	20.49	17.11	31.71	20.45	17.26	62.41
$L = 4$	Lena	24.14	9.62	122.14	**24.14**	**9.57**	**24.77**	24.00	9.75	31.35	24.01	9.77	219.14
	Cameraman	24.30	8.27	402.37	24.29	**8.22**	**46.92**	23.93	8.55	62.98	**24.47**	8.63	187.79
	Nimes	**26.37**	**8.16**	11.27	26.37	**8.16**	16.17	26.37	8.19	**6.43**	26.30	8.23	10.41
	Hdr	**23.29**	**12.15**	28.91	23.28	12.17	10.80	23.06	12.58	**4.41**	23.26	12.23	17.80
$L = 10$	Lena	**26.10**	7.53	71.72	26.09	**7.52**	**15.97**	25.97	7.54	18.95	25.82	7.95	124.40
	Cameraman	**25.97**	**6.92**	423.27	25.89	**6.99**	21.96	25.53	7.23	24.28	25.82	7.64	190.70
	Nimes	28.40	6.41	6.43	28.40	6.40	5.00	28.35	6.46	3.97	28.32	6.43	5.27
	Hdr	**25.32**	9.58	13.09	25.31	**9.57**	**6.94**	25.13	9.79	7.91	25.29	9.64	13.72

Bold values Indicate the best results in the comparison experiment

Table 2, the values of α and β are almost invariant in EFDM, FED and AOS. But for KSS method, the values of α and β are quite different. The reason mainly lies in the compression of image range to $[0, 1]^2$ in KSS for the convenience of fast Fourier transform.

From Table 2, we also find that the parameters of our model are image-dependent. Fortunately, the ranges of the parameters in our model for all test images are not large and can be chosen as follows: For all images, we recommend $\alpha \in [0.5, 2.5]$, and $\beta \in [0.5, 2.5]$ for EFDM, FED and AOS. For KSS, the range is recommended as $\alpha \in [0.5, 2]$ and

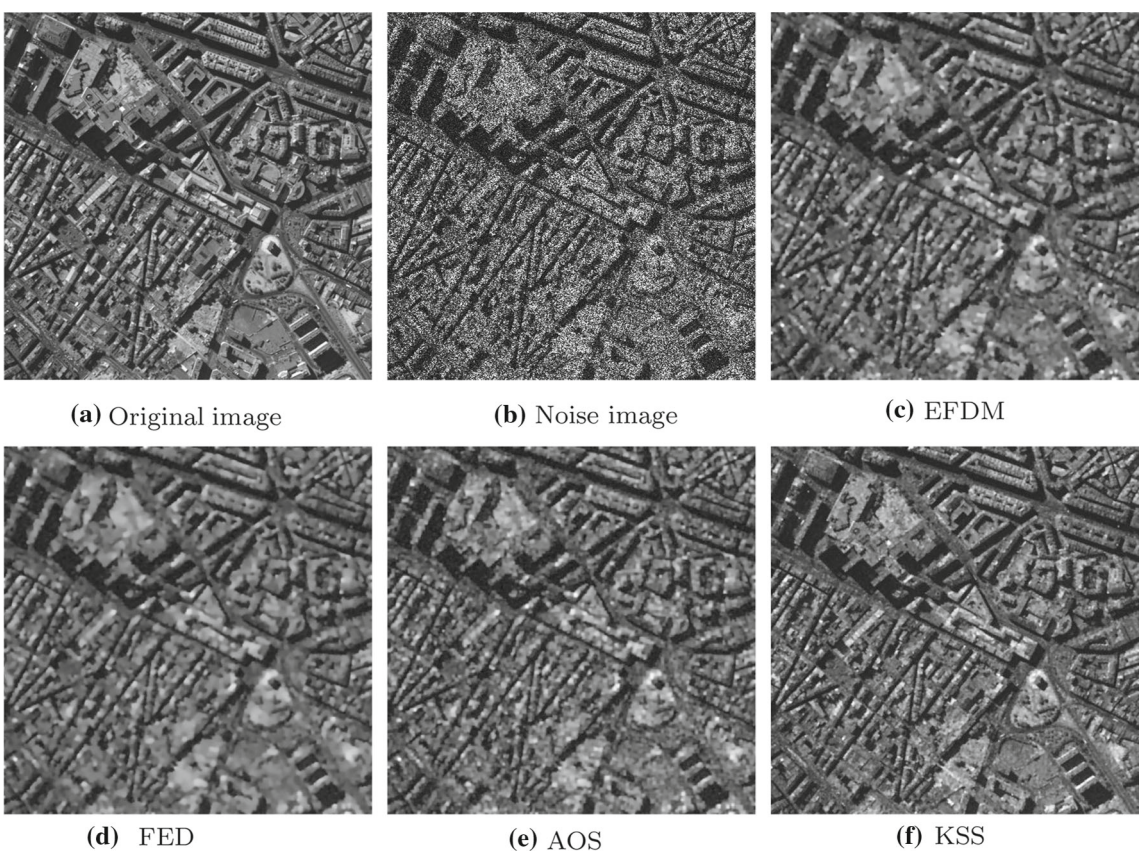

Fig. 2 Experimental results of algorithm 1–4 for Hdr with multiplicative Gamma noise $L = 4$

Table 2 The optimal β, α of four algorithms for our model, where $\sigma = 1$

		EFDM		FED		AOS		KSS	
		β	α	β	α	β	α	β	α
$L = 1$	Lena	1.7	1.7	1.6	1.7	1.7	1.8	1.3	1.5
	Cameraman	1.6	2	1.8	2	1.9	2.1	1	2
	Nimes	0.7	0.8	0.8	0.8	0.8	0.8	0.4	0.4
	Hdr	1	1.3	1	1.4	0.9	1.4	0.7	1.2
$L = 4$	Lena	1.5	2	1.5	2	1.5	2.4	1	1.2
	Cameraman	1.8	2.5	1.8	2.3	1.9	2.5	1.4	1.7
	Nimes	0.7	0.7	0.7	0.7	0.7	0.7	0.8	0.9
	Hdr	0.8	1.1	0.8	1.1	0.7	1	0.7	1.1
$L = 10$	Lena	1.5	2.2	1.6	2.2	1.7	2.6	1.3	1.6
	Cameraman	1.8	2.6	1.7	2.2	1.9	2.4	1.5	1.9
	Nimes	0.7	0.6	0.7	0.6	0.7	0.7	0.7	0.8
	Hdr	0.7	1	0.7	0.9	0.7	0.9	0.7	1

$\beta \in [0.5, 1.5]$. Our main purpose is to apply this denoising model to remove multiplicative Gamma noise, which usually occurs in satellite images such as SAR images. Therefore, proper parameters can be chosen based on the above parameter ranges in practical applications, and the results are quite acceptable and really close to the optimal ones.

5.3 Comparisons with Other Models

In this section, in order to demonstrate the ability of our method to deal with multiplicative Gamma noise, we test three images including the Synthetic image (256×256), Cameraman (256×256) image and Nimes (400×400) image that are contaminated by multiplicative Gamma noise with mean equals to 1 and $L \in \{1, 4, 10\}$. The proposed model is compared with AA, OS and DD model. AA and OS are two well-known variational methods that have desirable experimental results in removing multiplicative noise. Since our diffusion model is inspired by the DD model, it is necessary for us to make a comparison.

In numerical aspects, the numerical method for solving AA model is finite difference method based on the gradient descent approach and OS model is solved by nonlinear inverse scale space flow method proposed by Osher and Shi [37] in 2008. FED method is used to implement the DD model. All experimental results are obtained optimally by adjusting the parameters in each denoising method.

Figure 3 shows the results for the Synthetic image corrupted by multiplicative Gamma noise with $L = 1$. The Synthetic image is a challenge for all the models, since it contains gradients, edges and corners. As illustrated in Fig. 3, the

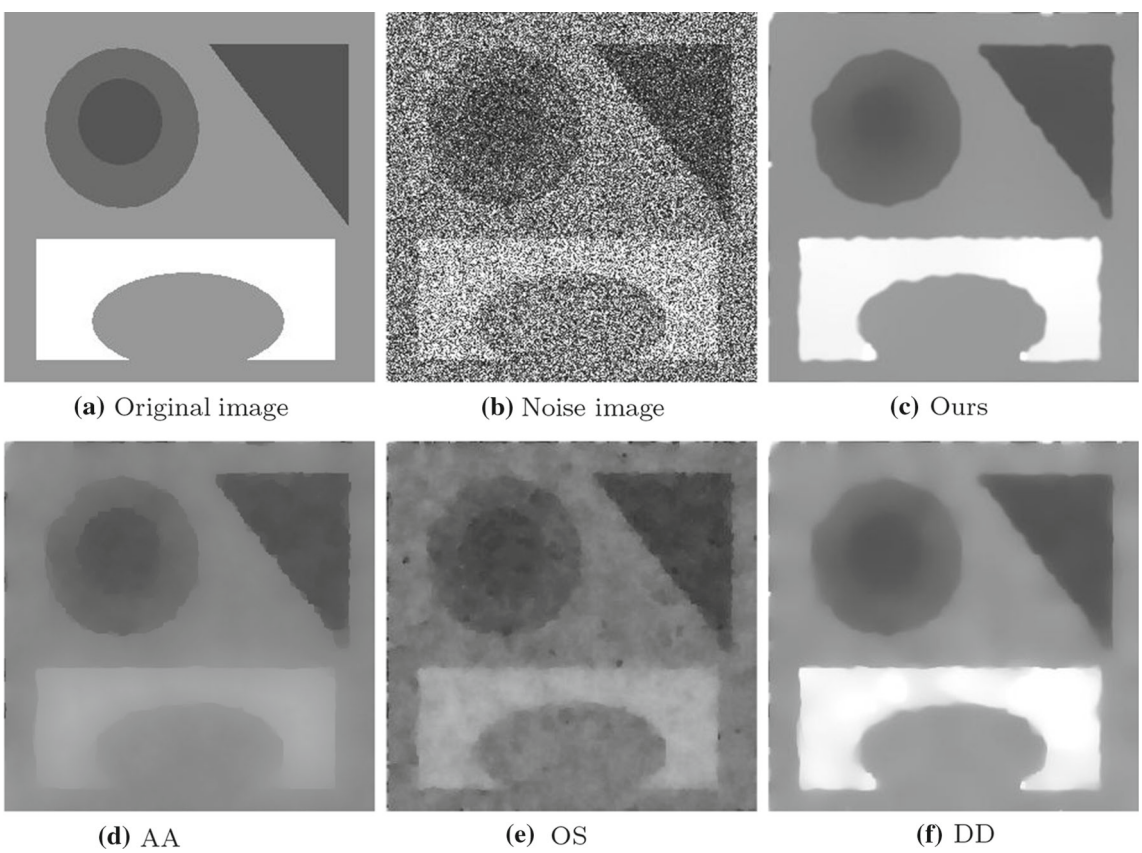

Fig. 3 Experimental results of AA, OS, DD and our model for the Synthetic image with multiplicative Gamma noise $L = 1$

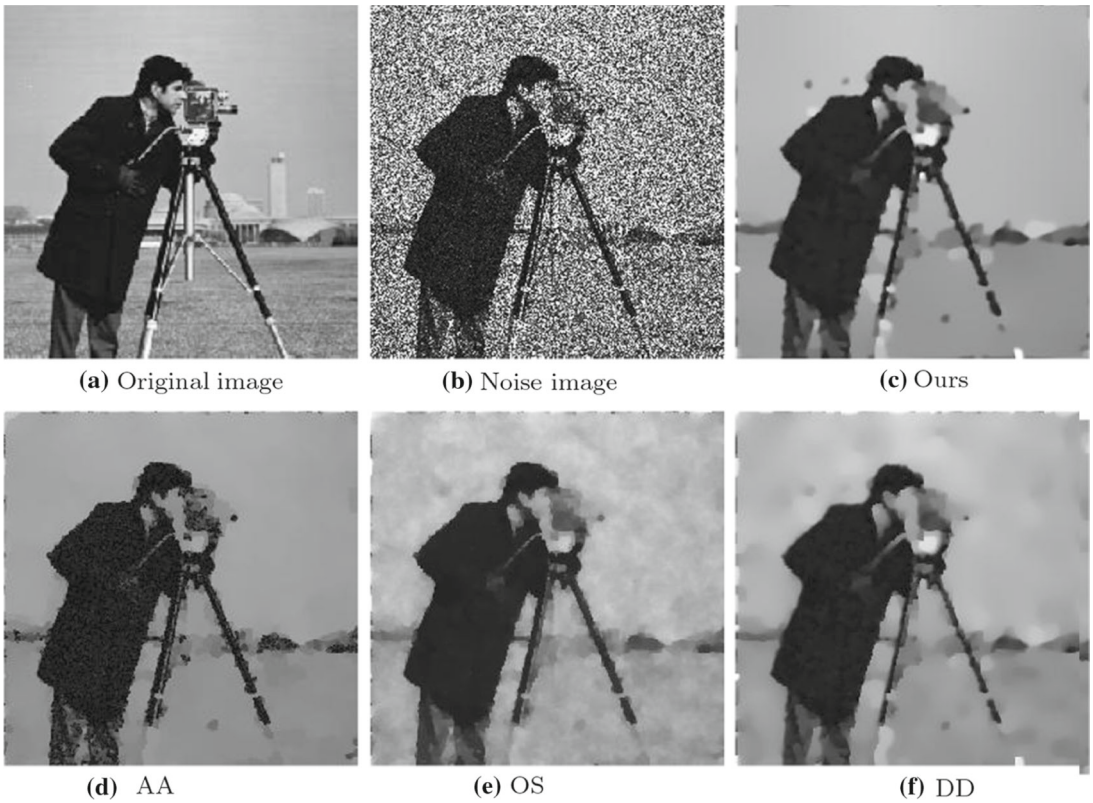

Fig. 4 Experimental results of AA, OS, DD and our model for the Cameraman with multiplicative Gamma noise $L = 1$

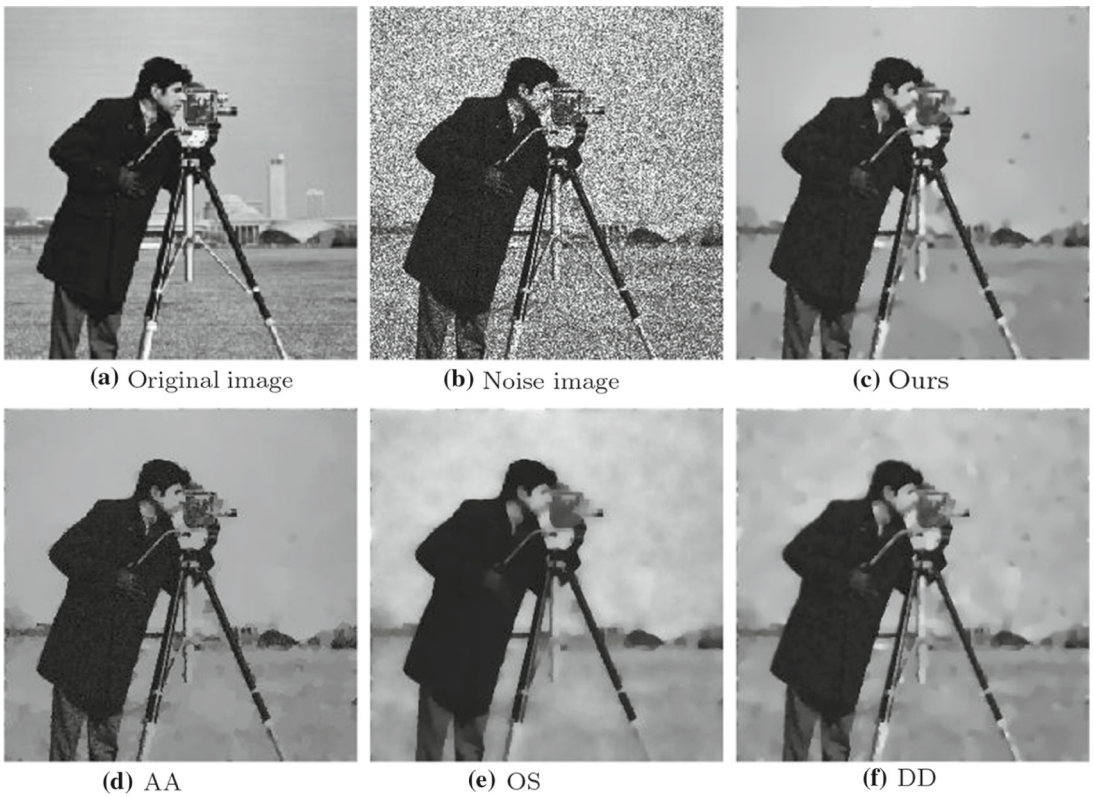

Fig. 5 Experimental results of AA, OS, DD and our model for the Cameraman with multiplicative Gamma noise $L = 4$

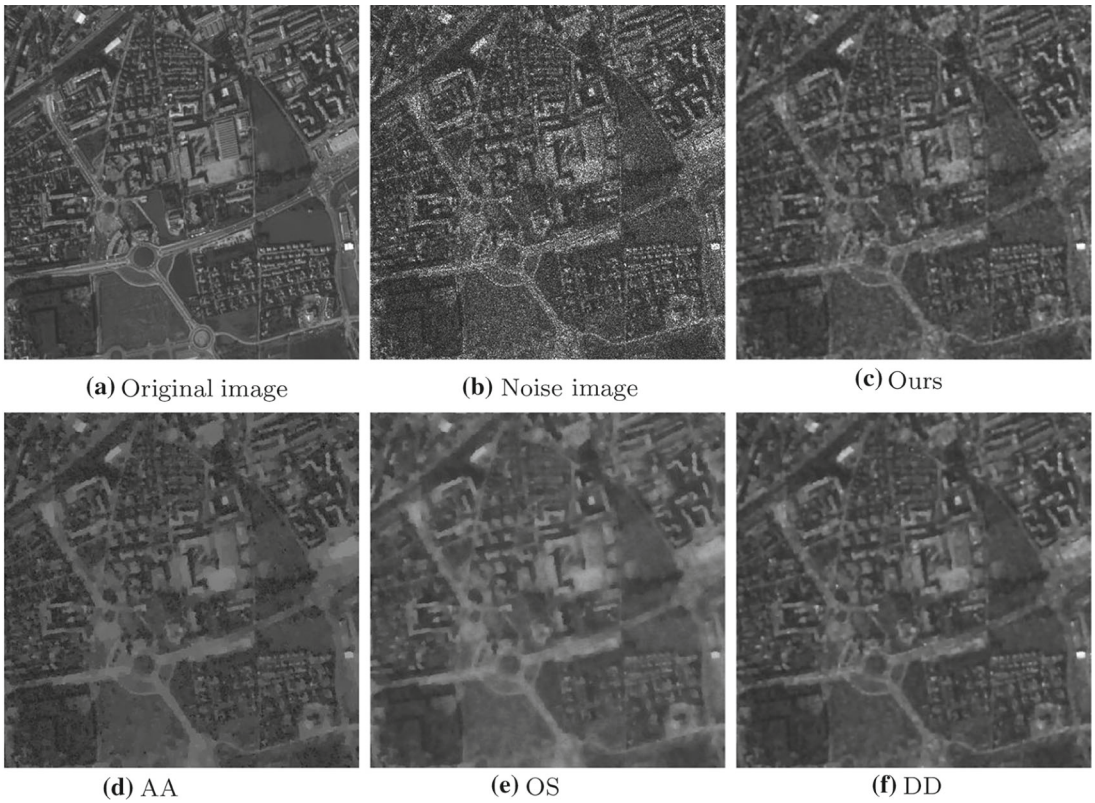

Fig. 6 Experimental results of AA, OS, DD and our model for the Nimes with multiplicative Gamma noise $L = 4$

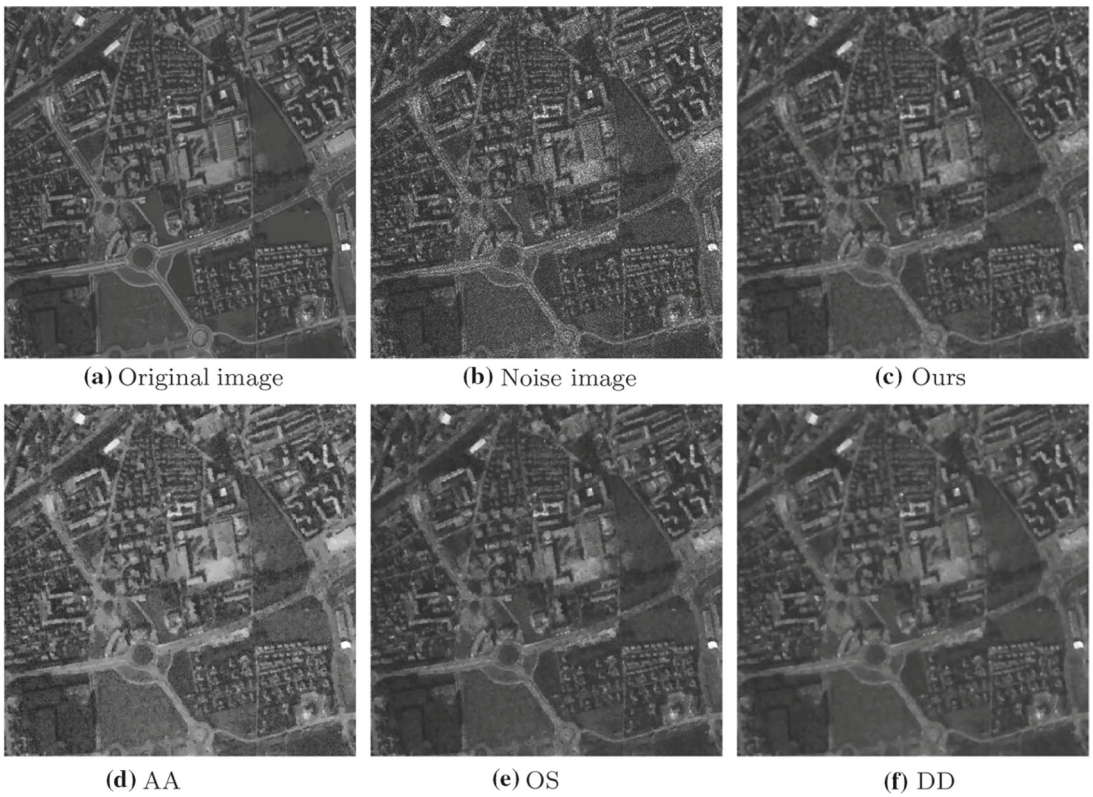

Fig. 7 Experimental results of AA, OS, DD and our model for the Nimes with multiplicative Gamma noise $L = 10$

Table 3 Comparisons of AA, OS, DD and our model, where $\sigma = 1$

		Ours		AA		OS		DD	
		PSNR	MAE	PSNR	MAE	PSNR	MAE	PSNR	MAE
$L = 1$	Synthetic	**24.31**	**5.56**	12.46	31.04	13.27	27.43	23.34	7.95
	Cameraman	**21.86**	**10.74**	16.49	29.50	21.00	13.41	20.93	12.85
	Nimes	**23.84**	**11.06**	22.04	12.60	23.44	11.24	23.80	11.10
$L = 4$	Synthetic	**28.24**	**2.62**	17.90	14.37	24.03	8.78	26.54	5.04
	Cameraman	**24.30**	**8.27**	22.21	12.98	22.45	10.56	23.60	9.63
	Nimes	26.37	8.16	24.52	9.58	25.03	9.17	**26.61**	**5.00**
$L = 10$	Synthetic	**30.85**	**1.88**	21.52	9.60	26.19	7.58	29.20	3.60
	Cameraman	**25.91**	**6.84**	25.20	8.08	24.80	9.08	25.50	7.97
	Nimes	**28.40**	**6.40**	26.80	7.58	27.59	6.94	28.13	6.55

Bold values Indicate the best results in the comparison experiment

restored results of AA and OS model are not very good when dealing with $L = 1$ case. AA model's result suffers from the unclear edges, and the restored image from OS tends to be piecewise constant. Even though the PSNR and MAE values of OS are much better than AA, they both change the mean gray value of the image. Our model and DD model have no such problem and obtain good results for the Synthetic image. It is worth noting that our model can protect edges of the image better than DD in the visual aspect.

Figures 4 and 5 show the comparisons of experimental results for the Cameraman image that contaminated by multiplicative noise with $L = 1$ and 4. For $L = 1$, our method can get the best results in the visual aspect. The visual quality of the restored image of our model is quite acceptable, especially for the background of Cameraman when $L = 4$ in Fig. 5, but our model does lose some fine boundaries when compared with AA model. Figures 6 and 7 show the results of Nimes for each model. Noise level are chosen to be $L = 4$ and $L = 10$. As we can see, when the noise level is low, the proposed model can remove noise and protect finer edge details, while AA has the black spots that should not appear.

Table 3 shows comparisons of our model with the AA, OS and DD by PSNR and MAE. When $L = 4$ and 10, all the models perform well in terms of PSNR and MAE. However, in $L = 1$ case, the diffusion equation-based models do better jobs in removing multiplicative noise than variational models AA and OS. Thus, we conclude that the fidelity term of variational model may not play a functional role when multiplicative noise level is high. In addition, our model performs better in terms of PSNR and MAE when compared with DD model, due to the different diffusion coefficient used in our model.

6 Conclusion

In this paper, a nonlinear smooth diffusion model is proposed to remove multiplicative Gamma noise. The mathematical analysis of the model has been carried out, such as the existence of the solution. Furthermore, experimental results show that the smooth diffusion model achieves better experimental results than others. Comparisons of four numerical methods provide that FED is the most powerful method in denoising speed and denoised performance and KSS method protects more details of the image than other methods in visual aspect. The mathematical properties of the KSS method for our model will be the future work.

Acknowledgements The authors would like to thank the reviewers for the valuable suggestions to the paper and professor Patrick Guidotti from the University of California, Irvine, for sharing his code. This work is partially supported by the Natural Science Foundation of Heilongjiang Province of China (Grant Nos. LC2018001, A2016003), the National Science Foundation of China (Grant Nos. U1637208, 51476047), the Natural Science Foundation (Grant No. 11871133), the Fundamental Research Funds for the Central Universities and Program for Innovation Research of Science in Harbin Institute of Technology (PIRS OF HIT 201609), and the Fundamental Research Funds for the Central Universities and Program for Innovation Research of Science in Harbin Institute of Technology (PIRS OF HIT 201601).

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